

Linux Commands Cheatsheet

High-level important commands — Ponmudi M R

Navigation & Files

`pwd / ls -la / cd /path`

→ Current dir / list all / change dir

`touch file.txt`

→ Create empty file

`mkdir folder / rm -rf folder`

→ Create / delete folder

`cp src dest / mv src dest`

→ Copy / move or rename

`cat file / less file / tail -f log`

→ View / scroll / follow live log

`find / -name '*.conf'`

→ Find files by name

`grep -r 'text' .`

→ Search text recursively in all files

Disk & Hardware

`lsblk / lsblk -f`

→ List disks / with filesystem & UUID

`df -h / du -sh folder/`

→ Free space / folder size

`sudo fdisk -l / blkid`

→ Partition table / UUID of partitions

`sudo mount /dev/sdb1 /mnt/usb / sudo umount /mnt/usb`

→ Mount / unmount device

`sudo smartctl -a /dev/sda`

→ Disk SMART health check

`lsusb / lspci / lscpu`

→ USB / PCI / CPU hardware info

`dmesg -w`

→ Live kernel log — watch USB plug events

`sudo lshw -short`

→ Full hardware summary

`sensors / nvidia-smi`

→ Temperature sensors / GPU stats

Format USB Drive

`lsblk`

→ Identify USB — check RM=1 and size

`sudo umount -l /dev/sda1`

→ Force unmount (if busy)

`sudo wipefs -a /dev/sda`

→ Wipe all filesystem signatures

`sudo parted /dev/sda --script mklabel msdos mkpart primary fat32 0% 100%`

→ Create new FAT32 partition

`sudo mkfs.vfat -F32 -n MUSIC /dev/sda1`

→ Format as FAT32 with label MUSIC

`lsblk -f`

→ Verify format success

■ System & Services (systemd)

`systemctl status ssh`

→ Check service status

`sudo systemctl start/stop/restart nginx`

→ Start / stop / restart service

`sudo systemctl enable nginx`

→ Auto-start at boot

`journalctl -u ssh -f`

→ Live service logs

`journalctl -b -1`

→ Previous boot logs

`top / htop / ps aux | grep nginx`

→ Running processes

`kill -9 PID`

→ Force kill a process

`df -h / free -h`

→ Disk / RAM usage

■ Networking

`ip a / ip r`

→ IP addresses / routing table

`ping -c 3 8.8.8.8`

→ Connectivity test

`ss -tulnp`

→ Open ports and listening services

`nmcli con show / nmcli dev status`

→ NetworkManager connections

`nmcli con mod 'WiFi' ipv4.dns '8.8.8.8 1.1.1.1'`

→ Set static DNS

`sudo systemctl restart systemd-resolved`

→ Restart DNS resolver

`resolvectl status`

→ Check active DNS servers

■ SSH & File Transfer

`ssh user@192.168.1.10`

→ Connect to remote machine

`ssh-keygen -t ed25519 -C 'email@example.com'`

→ Generate SSH key pair

`cat ~/.ssh/id_ed25519.pub`

→ View public key (add to GitHub)

`ssh -T git@github.com`

→ Test GitHub SSH connection

`scp file user@host:/path/`

→ Copy file over SSH

`rsync -avz src/ user@host:/dest/`

→ Sync folder over SSH (better than scp)

■ Git & GitHub

`git init / git clone git@github.com:user/repo.git`

→ Init / clone repo

`git remote set-url origin git@github.com:user/repo.git`

→ Switch remote to SSH

`git status / git diff`

→ Check changes

`git add . / git add file.txt`

→ Stage all / specific file

`git commit -m 'message'`

→ Commit staged changes

`git push origin main / git pull`

→ Push / pull from GitHub

`git log --oneline / git branch`

→ Commit history / branches

`git checkout -b new-feature`

→ Create and switch to new branch

■ Fedora / DNF Package Manager

`sudo dnf update`

→ Update all packages

`sudo dnf install package / sudo dnf remove package`

→ Install / remove

`sudo dnf search vlc / dnf list installed`

→ Search / list installed

`sudo rpm -ivh package.rpm / sudo dnf install ./pkg.rpm`

→ Install .rpm file

`flatpak install flathub org.videolan.VLC`

→ Install via Flatpak

`flatpak list / flatpak update`

→ List / update Flatpak apps

■ Permissions & Users

`chmod +x script.sh / chmod 755 file`

→ Make executable / set permissions

`chown user:group file`

→ Change owner

`sudo adduser ponmudi / sudo passwd ponmudi`

→ Add user / set password

`sudo usermod -aG sudo ponmudi`

→ Add to sudo group (Ubuntu)

`sudo usermod -aG wheel ponmudi`

→ Add to sudo group (Fedora)

`whoami / who / id`

→ Current user / logged-in / user info

■ Text Processing

`cat file / nano file / gedit file`

→ View / edit in terminal / edit in GUI

`echo 'text' > file / echo 'text' >> file`

→ Write / append to file

`grep 'text' file / grep -r 'text' .`

→ Search in file / recursively

`sed 's/old/new/g' file`

→ Replace text in file

```
awk '{print $1}' file / cut -d',' -f2 file
```

→ Print column / cut CSV field

```
wc -l file / sort file | uniq
```

→ Count lines / sort + deduplicate

```
diff file1 file2
```

→ Compare two files

■ Terminal Customization (.bashrc)

```
nano ~/.bashrc / source ~/.bashrc
```

→ Edit / reload bashrc

```
alias ll='ls -la' / alias c='clear'
```

→ Create shortcuts

```
echo -e "\e[1;35m$(figlet -c -t -f big PONMUDI)\e[0m"
```

→ Violet centered ASCII banner

```
PS1='\[\e[1;32m\]\u@\h\[\e[0m\]:\[\e[1;34m\]\w\[\e[0m\]\$ '
```

→ Custom colored prompt

```
sudo apt install figlet toilet lolcat neofetch
```

→ Install terminal decoration tools

```
neofetch
```

→ Show system info with OS logo